



ME 218c Spring 2026: Karlival of the Animals Navigate the Fountain to Collect the Animals

Grading Session on May 26 from 11-5 pm.

Project Presentation on May 27 starting at 5:00 pm.

Goal:

The goal of this project is to provide a framework in which you can apply your knowledge of microcontrollers and multi-processor communications to a task that will provide an enjoyable experience for users and observers alike!

Purpose:

The underlying purpose of this project is to provide you with an opportunity to integrate all that you have learned in the ME218 course sequence, with an emphasis on the new material in ME218c.

Background:

After restoring power to the coal factory and rebuilding your time machine, you attempt to jump back to the present. Instead, the machine sputters, hisses, and crashes into an entirely new district of the Karlival: the Grand Hydraulic Gardens. Here, towering fountains and steam-fed waterways power the city's newest attractions.

Crowds gather around the famed Royal Fountain Arena, where swirling currents carry prized fountain creatures—ducklings, turtles, fish, and other curious animals. Guiding each vessel is a legendary **Captain Duck**: a battle-tested waterfowl known for its fearless swagger and unmistakable style. It is said no vessel can claim victory without a Captain Duck aboard to command the waters.

The Task:

Your task is to design and build a **Steampunk-Themed Remote Controlled Watercraft** that can navigate Terman fountain. Your vessel must blend into the Karlival aesthetic while demonstrating modern mechatronic capability.

Your watercraft must maneuver through the fountain, herd and collect as many animals as possible from the moving currents, and strategically secure them for points while protecting the Captain Ducks riding aboard your craft. If your captains are knocked off their **DuckPlate** or lost to the water, your crew loses both guidance and glory.

The system should integrate sensors, actuators, propulsion, and event-driven software running in C on a PIC32 microcontroller. A successful design will be both visually steampunk and technically sophisticated, showcasing aquatic mobility, intelligent strategy, and precise control in a dynamic hydraulic arena.

Specifications

General:

- ☐ Each team will construct a **Quackraft** and a **Mallard Module**.
- ☐ The Mallard Modules are I/O devices which control various aspects of the Quackraft functions, and will contain an SPDL supplied XBee radio module to communicate wirelessly between Mallard Modules and Quackrafts.
- ☐ The Quackrafts are watercraft that can be wirelessly driven around the arena.
- ☐ The class will form a Communications Committee to draft a class-wide standard communications protocol that will permit any Mallard Module to effectively control any Quackraft with which it is paired.

Fauna:

- ☐ Fauna are approximately 50mm x 50mm objects with the properties of rubber bath toys. Due to different fauna species and inconsistent process control across fauna manufacturing, **significant variations should be expected**.

- ☐ All fauna will be supplied by the teaching staff.
- ☐ Fauna is generally buoyant, meaning it will likely float at the surface of the water.
- ☐ Fauna shall not be modified in any way.

The Mallard Module:

- ☐ The Mallard Module is a controller capable of pairing with **any** Quackraft.
- ☐ The Mallard Module shall contain all controls and displays relevant to propulsion, fauna collecting functionalities, charge status, recharging, and wireless communication with the Quackraft (these are each elaborated on below).
- ☐ Quackrafts require steam pressure to function. **Thus, the Quackrafts cannot do any work when there is no steam remaining (See [The Quackraft](#) for a more thorough explanation of this).**
- ☐ The Mallard Module shall have two modes: Drive Mode and Refuel Mode. It **must be impossible** to be in both modes simultaneously.
 - ☐ **Drive Mode**
 1. When in Drive Mode, the Mallard Module shall be capable of controlling the propulsion.
 2. Any recharging input shall be ignored while in this state.
 - ☐ **Refuel Mode**
 1. When in Refuel Mode, the Mallard Module shall be allowed to increase the steam pressure of the Quackraft.
 2. Any driving or mechanism control inputs shall be ignored while in this state.
 3. While in this state, refueling should only occur while the user of the Mallard Module partakes in some sort of refueling action.
 4. The refueling action of the Mallard Module must implement an activity consisting of large-scale human body movements. It should be inventive and interesting for the audience to watch. Use of actions that make the Mallard Module operator look and feel foolish are encouraged.
 5. The required Mallard Module refueling action must last continuously for at least 5 seconds. Partial completion of a refueling action should result in partial refueling of the Quackraft, proportional to time spent performing the refueling action.
- ☐ The Mallard Module must include, at minimum, an indicator to display the remaining charge level in a Quackraft with a resolution of 1 part in 5.
- ☐ The Mallard Module shall have some method of displaying whether or not it is paired with a Quackraft. If a Mallard Module is paired with a Quackraft, it shall have some method of indicating which Quackraft it is paired to.
- ☐ Mallard Modules must incorporate at least 1 analog input and at least 1 electromechanical output.
- ☐ Each Mallard Module shall contain a SPDL-issued XBee radio module used to communicate with a Quackraft.
- ☐ The Mallard Module must provide the user controls for all required functions determined in the class-wide communications protocol.
- ☐ Each Mallard Module must implement the class-wide protocol for coordinating game information (See [Communications](#)).
- ☐ Each Mallard Module may only be powered by 5V LiIon battery packs.
- ☐ The size, shape and mass of the Mallard Module are constrained only to what is portable by your team. Bear in mind that you may need to move the Mallard Module to stay in radio range with a Quackraft.

The Quackraft:

- ☐ Each Quackraft is a watercraft capable of operating in the fountain.
- ☐ Each Quackraft shall be equipped with a propulsion/maneuvering system controlled by the Mallard Module.
- ☐ The only component of the Quackraft that may extend outside the outer edge of the bumper is a 12 inch length of foam tubing provided for the purpose of dethroning CaptainDucks by the teaching team.
- ☐ The **DuckPlate must be 8in +/- 1" above the water line** and one edge must be tangent to the interior edge of the bumper.
- ☐ Each DuckPlate will have a 9" radius non-interfering zone. No part of the Quackraft should interfere in this region.
- ☐ Each DuckPlate will be supplied by the teaching team and hold a maximum of 4 Captain Ducks.
- ☐ You can glue to the bottom of the DuckPlate. Do not modify the surface of the top of the DuckPlate.

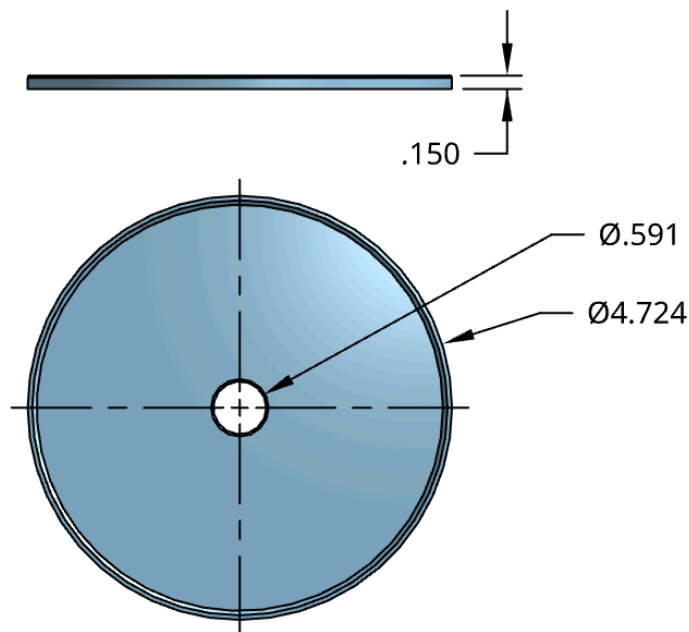


Figure 1: The DuckPlate, dimensions in inches.

- ☐ Each Quackraft shall display the number of the team that built it with digits at least 15cm tall.
- ☐ Each Quackraft shall have around its largest perimeter two SPDL-issued closed-cell foam bumpers. The two 1.75 inch diameter foam bumpers must be stacked directly on top of each other. The bottom edge of the bottom bumper must be at least 2 inches above the waterline. The top edge of the top bumper must be less than 7.5 inches above the waterline.
- ☐ Each Quackraft is limited to a total waterline circumference of 6 ft.
- ☐ No part of the Quackraft may protrude beyond the circumference of the foam bumper, with exception of any implement designed to dethrone captain ducks.
- ☐ Collisions between Quackrafts will be unavoidable. Ensure that your Quackraft is designed to robustly absorb crash energy at the bumper.
- ☐ Each Quackraft must implement an electromechanical indication of its state of association with a Mal-

lard Module. Exactly which association states must be displayed is at the discretion of the Communications Committee.



Figure 2: The Captain Duck...a creature with fearless swagger and unmistakable style.

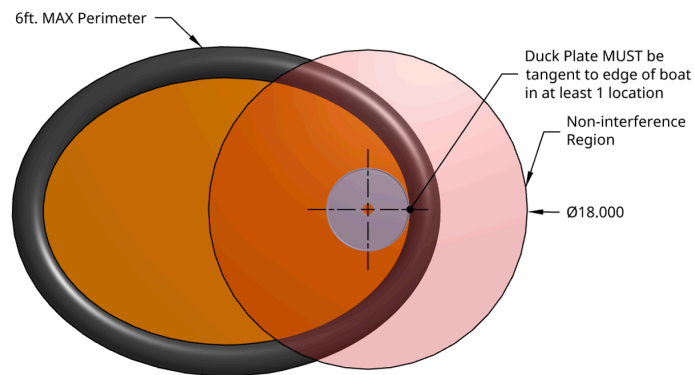


Figure 3: Quackraft Top View, dimensions in inches

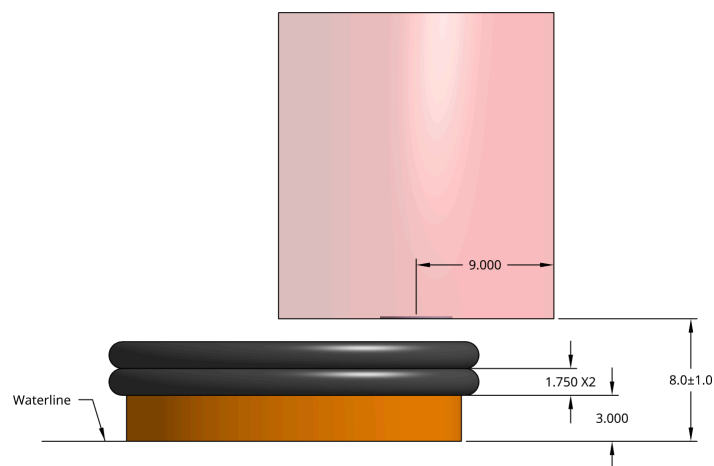


Figure 4: Quackraft Side View, dimensions in inches

- ☐ Each Quackraft must keep track of its own steam pressure status, counted in seconds. Quackrafts can hold a maximum of **40** seconds worth of steam.
- ☐ Each action that a Quackraft takes that propels it, deducts steam pressure for the duration of the action. Using any actuators designed to de-throne Captain Ducks also deducts steam for the duration of the action. Using propulsion and de-throning actuators at the same time uses up steam power at double time. A Quackraft that is sitting idle, whether paired with a Mallard Module or not, is not considered to be actively consuming steam.
- ☐ **Once there is zero steam pressure remaining, a Quackraft cannot activate its propulsion or any mechanisms until refueled through a refueling action from the Mallard Module.**
- ☐ Each Quackraft may only be powered by SPDL-supplied 7.2V NiMH and 5V LiIon battery packs. Up to 2x 7.2V NiMH packs and 1x 5V LiIon pack may be used.
- ☐ The enclosure of each Quackraft's sensitive instrumentation and propulsion systems must be protected against damage from ingress of objects and water.
- ☐ Each Quackraft shall contain a SPDL-issued XBee radio module used to communicate with a Mallard Module.
- ☐ Each Quackraft must implement the class-wide protocol for coordinating game information (See Communications).

Basic Game Play:

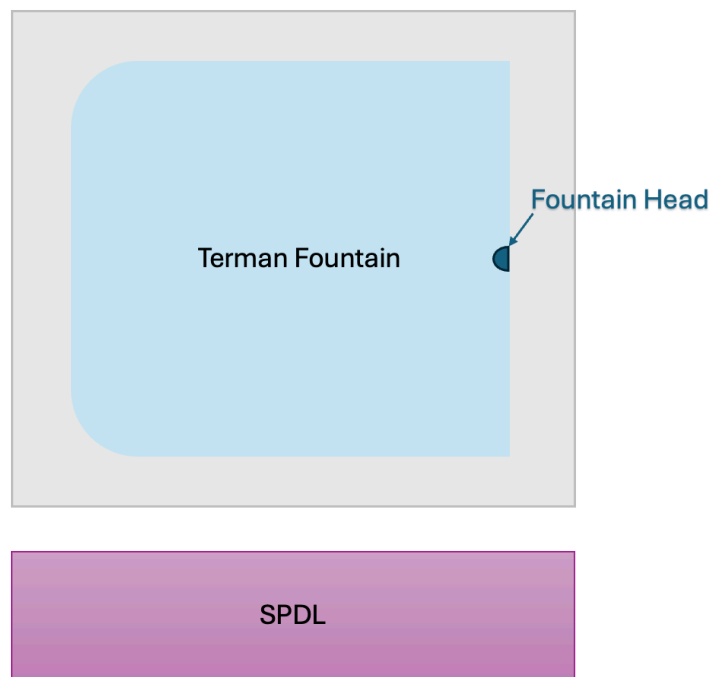


Figure 5: The Royal Fountain's Game Play Region.

- ☐ At the beginning of each game, the participating Quackrafts will be paired to Mallard Modules according to pairs decided by the teaching team.
- ☐ Each Quackraft will start by holding 4 Captain Ducks and with a full tank of steam.
- ☐ A game round proceeds in real time, with no turn-taking.

- ☐ A game round consists of each Quackraft continually racing through the fountain described above to collect fauna and dethrone opposing teams' captain ducks.
- ☐ Any team caught removing their boat from the water/wading into the water to replace captain ducks will be considered dead/disqualified for the rest of the game round.
- ☐ Between the five teams, there will be a fixed amount of fauna.
- ☐ A game round ends after six minutes.
- ☐ Every fauna that is within the perimeter of a Quackraft at the end of the game counts for 1 point. Opposing CaptainDucks count as 1 point like any other fauna if captured.
- ☐ Every Captain Duck that is still on its DuckPlate at the end of the game counts for 4 points.
- ☐ Playing defense or ramming another team's Quackraft is acceptable.

Communications:

- ☐ Communications between Quackrafts and Mallard Modules will take place over the airwaves using SPDL-supplied XBee radio modules in API mode.
- ☐ Each Mallard Module and Quackraft **shall** communicate with the XBee over an asynchronous communications channel using 9600 baud, 8N1 at 3.3V levels.
- ☐ Messaging between Mallard Modules and Quackrafts is limited to 5 Hz. That is to say that a single Mallard Module shall transmit one message every 0.2 s, with the paired Quackraft transmitting one message also during this period.
- ☐ Because of the inherent unpredictability in wireless latency, Mallard Modules and Quackrafts must be able to accept any message at any time, and may not have a fixed time window in which they are open to message reception.
- ☐ Human inputs, such as button presses and direction changes, may only take effect on the following transmitted message. Extra messages (and thus a greater than 5 Hz message rate) may not be generated as a result of a human action.
- ☐ Any other hardware or implementation requirements or recommended practices are left to the Communications Committee.
- ☐ The details of the communications protocol will be defined and specified by a Communications Committee, which will consist of a designated representative from each team. The specification must be in a written form and with sufficient detail that someone skilled in ME218 material could implement it.
- ☐ The class communications protocol must be defined to support the functional requirements listed earlier in this document. The Communications Committee is free to write a protocol of any complexity that fulfills the functional requirements. If a particularly clever messaging definition reduces overhead while maintaining the required functionality, this is perfectly acceptable. Or, if the Communications Committee implements a superset of the functionally required messaging, that would also pass.
- ☐ The communication protocol must define any addressing and packet formats if required.¹
- ☐ The communication protocol shall cover all communication handled through the XBee, including pairing, operation, unpairing, and exception handling between a Quackraft and a Mallard Module. Interruptions in wireless communication are frequent and occur at irregular intervals, and the protocol should include some robustness against such interruptions.
- ☐ While a clear division of labor is not obvious, we strongly encourage making an effort to have all team members implement some portion of the communications in software (as opposed to just the member on the Communications Committee).

¹That is, Layer 3 of the OSI model.

General Requirements:

- ☐ Automation of the setting of any Mallard Module input is prohibited. That is to say that an algorithm shall not set thrust magnitude, direction, etc. An algorithm may advise the human operator of a Mallard Module how to set the various controls to achieve a desired outcome, however, in the end it is the human who must dial the setting, using the inputs provided, that generates a measurable effect in the Quackraft.
- ☐ There is no class-imposed upper limit on the number of processors employed; however, you must use only the PIC32MX170F256B or PIC10F322. Tivas, Arduinos, Raspberry Pis, Teensys, Jetsons, and other microcontrollers or processors are not permitted. Clockwork computers are allowed.
- ☐ You are limited to an expenditure of **\$200.00/team** for all materials and parts used in the construction of your project. Materials supplied to each team by SPDL, from the lab kit, or the Cabinet Of Freedom do not count against the limit. All other items count at their fair market value. If it's an issue with something from the kit, we will provide a replacement for free, but we can't guarantee latency. Be careful with your components.
- ☐ A project logbook must be maintained for each group. A blog is appropriate to meet this requirement as long as it is made available to the teaching staff for review. This log should reflect the current state of the project, planning for the future, results of meetings, designs as they evolve, etc. The project logbook will be reviewed at irregular intervals for evaluation. The logbook can help improve communication within the team and help when writing the report.
- ☐ A report describing the technical details of the system will be required. The report should be of sufficient detail that a person skilled at the level of ME218c could understand, reproduce, and modify the design. The report must be in website format, and be suitable for posting on the SPDL site.
- ☐ Mallard Modules or Quackrafts based substantially on purchased platforms are not allowed.
- ☐ All projects must respect the spirit of the rules. If your team is considering anything that may violate the spirit of the rules, you must consult a member of the teaching staff.

Safety:

- ☐ The Mallard Modules should be safe, both to the user and the spectators.
- ☐ Intentionally ramming other Quackrafts is encouraged. However, prohibited actions include, but are not limited to, fouling the propulsion systems of Quackrafts, and/or jamming communications between Mallard Modules and Quackrafts. Unless it's raspberry.
- ☐ No part of the Quackraft may become ballistic. It is ok for Quackrafts to cause water to become ballistic.
- ☐ There have been no proven negative health effects due to radiation from XBee networks.
- ☐ The teaching staff reserves the right to disqualify any device considered unsafe.

Checkpoints

Design Review:

On 5/5/26 we will conduct a design review in lecture, one team at a time. Each team should prepare a few images showing your proposed designs for the Quackrafts. You will have 5 minutes to walk us through your ideas. **The focus should be on system level concepts², not detailed hardware or software.** We will spend the balance of the time giving feedback and asking questions. In addition to your concepts, you must present, as a PDF, your plan for the development, integration and testing steps that you will follow to complete the project. The plan must identify what functionality you will demonstrate at the three check-points along with the test procedures that you will use to prove that your team has met the checkpoint. Checkpoint tests must follow an incremental integration strategy with each successive checkpoint

²I/O, signal conditioning architecture, etc.

demonstrating all of the functionality of the prior checkpoint(s) as well as the new functionality. This plan must be approved by the teaching staff. If we feel that it is seriously flawed, we will ask you to revise and resubmit the following day. Lastly, each team must inform the teaching team of their representative for the Communications Committee during their design review.

"Splish-splash, I was taking a bath"

Bobby Darin, Splish Splash

First Draft of Communications Standard:

Due by 5:00 pm on **5/8/26**. This draft will be made available to the entire class, so that everyone is ready to deliver feedback at the in-class review.

"I'm the tidal wave of power to wash you away! Put out the fire, Crasher Wake!"

Crasher Wake, Brilliant Diamond and Shining Pearl

In-Class Communications Standard Review:

In class on **5/12/26** we will conduct a top-to-bottom review of the Communications Committee's draft protocol. Bring your prepared questions, concerns, and suggestions for improvement! Everyone should attend, if possible—the more eyes we can put on the protocol early, the earlier we can catch the weird edge cases.

"It is only in deep waters that a fish knows how strong it is."

Matshona Dhliwayo

Communications Standard Working Draft:

Due by 5:00 pm on **5/13/26**. This is the working draft of the communications standard.

"As for me, I am tormented with an everlasting itch for things remote. I love to sail forbidden seas, and land on barbarous coasts."

Herman Melville

Finalized Communications Standard:

The final working version of the communications standard is due by 5:00 pm on **5/15/26**. No further changes are allowed to the standard. This protocol will be evaluated with respect to its completeness and suitability for the proposed system.

"And if anybody asks, just say a flood is coming."

Evan Almighty

First Checkpoint:

On or before **5/16/26**, you must demonstrate your approved 1st checkpoint functionality according to your defined testing procedure. Note: this is a functional evaluation only. The focus should be on demonstrating functional hardware and software. You may submit for approval a final revision of your checkpoint plan at this time.

"Just keep swimming. Just keep swimming."

Dory

Second Checkpoint:

On or before **5/20/26**, you must demonstrate your approved 1st and 2nd checkpoint functionality according to your defined testing procedure. The functionality demonstrated at this time must include full implementation of the communications protocol.

"[barnacle noises]"

Barbaracle, Pokémon X and Y

Third Checkpoint:

On or before **5/23/26**, each team must demonstrate (in addition to the 1st & 2nd checkpoints' functionality) your approved 3rd checkpoint functionality. The functionality demonstrated at this time must include a demonstration of interaction with at least one other teams' Quackrafts in addition to your own. Note that this checkpoint can be completed inside the SPDL (it's the Grading Session that will be completed in the fountain).

"You're gonna need a bigger Quackraft."

Chief Brody, an alternate version of Jaws

Grading Session:

During the Grading Session on **5/26/26**, each team will be required to demonstrate the ability to successfully participate in a game. This will include

1. Pairing with, and successfully operating, at least one other Quackraft constructed by another team;
2. Demonstrating all required functionality of the Mallard Module, including user interface and implementation of the Communications Committee-designed communications protocol, including:
 - (a) All functionality of the Drive Mode.
 - (b) All functionality of the Recharge Mode.
3. Demonstrating all required functionality of the Quackraft, including propulsion, recharging, pairing, and error handling.
4. Demonstrating successful execution of at least sixty seconds of play.

A detailed grading check-off procedure will be published at a later date.

"When I left you I was but the learner, now I am the master."

Darth Vader

Public Presentation:

This will take place on **5/27/26** starting at 5:00 pm at the Flooded Warehouse (outside of Thornton). At this event, members of the public will be encouraged to watch you navigate the gardens to collect the animals.

Report:

Draft due on **6/1/26** by 4:00 pm. The final version (with revisions incorporated) is due by 5:00 pm on **6/5/26**.

Celebration:

A celebration of the past 3 quarters of ME218 will take place at the Alpine Inn on **06/12/2025** starting at 3:00 pm. Mark your calendars now and save the date.

Evaluation

Performance Testing Procedures:

Each team will demonstrate their Mallard Module and Quackraft during the three checkpoints. Members of the teaching team will randomly assign a Quackraft to each Mallard Module during the grading session.

Grading Criteria:

- ☐ **Concept (10%)** This will be based on the technical merit of the design and coding for the machine. Included in this grade will be evaluation of the appropriateness of the solution, as well as innovative hardware, software and use of physical principles in the solution.
- ☐ **Implementation (15%)** This will be based on the prototype displayed at the evaluation session. Included in this grade will be evaluation of the physical appearance of the prototype and quality of construction. We will not presume to judge true aesthetics, but will concentrate on craftsmanship and finished appearance.
- ☐ **First Checkpoint (10%)** Based on the results of the performance demonstrated on 5/16/26.
- ☐ **Second Checkpoint (10%)** Based on the results of the performance demonstrated on 5/20/26.
- ☐ **Third Checkpoint (10%)** Based on the results of the performance demonstrated on 5/23/26.
- ☐ **Performance (20%)** Based on the results of the performance testing during the Grading Session.
- ☐ **Report (10%)** This will be based on an evaluation of the report. It will be judged on clarity of explanations, completeness and appropriateness of the documentation.

- ☐ **Report Review (5 %)** These points will be awarded based on the thoroughness of your review of your partner team's report. Read the explanations, do they make sense? Review the circuits, do they look like they should work?
- ☐ **Log Book (5 %)** This will be evaluated by the evidence of consistent maintenance as well as the quality and relevance of the material in the log book.
- ☐ **Housekeeping (5 %)** Based on the timely return of SPDL components, cleanliness of group workstations as well as the overall cleanliness of the lab. No grades will be recorded for teams who have not returned all loaned materials.

Resources

Websites:

[SparkFun](#)
[Newark](#)
[DigiKey](#)

[Seeed Studio](#)
[Ponoko](#)
[McMaster-Carr](#)

[Jameco](#)
[Adafruit](#)
[HobbyKing](#)

[Mouser](#)
[Hackaday](#)
[ServoCity](#)

You may also find [PlantUML](#) and [PlantText](#) helpful for creating message sequence diagrams.

Local Stores:

J&M Hobby House in San Carlos
[Jameco](#) in Belmont
[TAP Plastics](#) in various locations

Gems of Wisdom:

Be sure to check out [The ME218 Archive](#) for guidance from past generations.

Keep in mind that there's still plenty to be doing while one of your teammates is getting the Comm Protocol sorted. Make effective use of this time to develop and test other systems.

Revision History

Revision 0: Initial (4/30/26)